

HAND IN

Answers recorded on exam paper

Page 1 of 16 pages

QUEEN'S UNIVERSITY FINAL EXAMINATION
FACULTY OF ENGINEERING
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING
SOFT / CISC 437 - PROFESSORS SAMIR MOHAMMAD
April 18, 2022

INSTRUCTIONS TO STUDENTS:

This examination is 3 HOURS in length.

There are 4 sections to this examination. Please answer all questions in the exam papers

The following aids are allowed:
Casio FX-991 calculator

STUDENT NUMBER

Before you start, put your student number on top of all odd pages of all answer exam papers, including the front.

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Before you start, put your student number on top of all odd pages of all answer exam papers, including the front.

- Keep your test papers stapled.
- There is a blank spaces and/or page(s) provided in the quiz for rough work.
- If you answer a question on a different page, write us a note telling us where to find your answer.
- Comments are not required since you're writing under time pressure.
- We will only mark one answer for each question. If you write a rough answer and then a final answer, cross out the rough work or circle the final answer to tell us which one you want us to mark.
- You may write your answers in pen or pencil but **do not write in red or pink or purple.**

PLEASE NOTE:

Proctors are unable to respond to queries about the interpretation of exam questions.

Do your best to answer exam questions as written.

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The following area is for marking purposes only:

1.	_____	/ 10
2.	_____	/ 10
3.	_____	/ 10
4.	_____	/ 20
Total	_____	/50

Question 1: True / False. Write your Answer at the provided space.
(10 Points, 1 points each, no partial point)

T / F

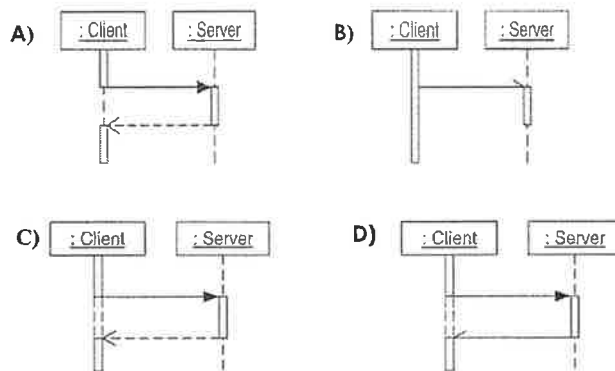
- 1.1) We manage performance in a software system after we build it according to a given specs and requirements. _____
- 1.2) SPE can be easily integrated into the software development processes, such as waterfall model, spiral model and rational unified process. _____
- 1.3) A use case is an instance of scenario. _____
- 1.4) Execution graphs are one type of software execution model with visual representation that helps to communicate execution behavior. _____
- 1.5) Closed QNM has external arrivals or departures. _____
- 1.6) Three reasons for supplementing the standard tools with instrumentation: convenience, data granularity, and control. _____
- 1.7) Patterns document common mistakes during development, also document solutions. _____
- 1.8) Parallel Processing Principle is related to execute processing in parallel (only) when the processing speedup offsets communication overhead and resource contention delay. _____
- 1.9) A refactoring is a correctness-preserving transformation that improves the quality of the software. _____
- 1.10) According to the One-Lane Bridge, one or only a few, processes may continue to execute concurrently. All other processes must wait. _____

Question 2: Multiple Choices. Circle the correct answer (A, B, C, or D).
(10 Points, 1 points each, no partial point)

2.1) _____ is the number of requests that can be processed in some specified time interval?

- A) Response time
- B) Throughput
- C) Responsiveness
- D) Scalability

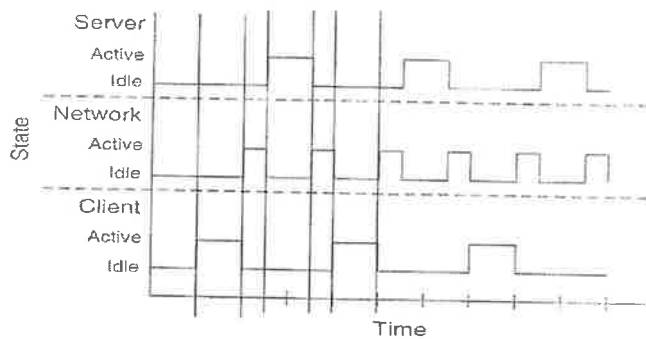
2.2) Which of the given figures represents synchronous communications.



2.3) A Pardo node represents:

- A) the steps that may be executed
- B) conditional execution of processing steps
- C) parallel execution within a scenario
- D) None of the above

2.4) The Figure below represents :



- A) Asynchronous Callbacks
- B) Deferred Synchronous Communication
- C) Synchronous Communication
- D) Asynchronous Communication

2.5) Factor(s) that may affect Software Performance Measurements is/are _____.

- A) Measurement Timing
- B) Reproducible Results
- C) System overhead
- D) All the above

2.6) You instrument software by inserting _____ at key points to measure pertinent execution characteristics.

- A) Stubs
- B) Code Probes
- C) Application program interface (API)
- D) All the above

2.7) The primary consideration(s) when designing instrumentation is/are:

- A) Defining the events to be measured
- B) Choosing the granularity of the measurements
- C) Dynamically selecting the data to be recorded
- D) All the above

2.8) Which Principle(s) for Performance-Oriented Design that are concerned with improving the overall performance of a system via cooperation among processes competing for computer resources.

- A) Performance Control principles
- B) Independent principles
- C) Synergistic principles
- D) All the above

2.9) _____ identify the dominant workload functions and minimize their processing.

- A) Centering principle
- B) Fixing-Point principle
- C) Locality principle
- D) Processing versus Frequency Principle

2.10) To avoid corrupting the data, during an update of a bank account records, we require exclusive access until the record has been updated. There are many options to do so. Which option that leads to minimizing the scheduling time and minimizing holding time.

- A) Lock the entire database
- B) Lock only the individual records
- C) Lock a group of records
- D) None of the above

Question 3: Short answer Questions (10 points, 1 points each, possibility of partial mark)

Please make your answers are brief, very precise to the point to get full mark.

3.1) Define SPE.

3.2) One of the steps of SPE process for Object-Oriented system is to verify and validate the models.
What are model verification and validation.

3.3) The UML allows you to specify timing requirements through the use of timing marks, time expressions timing constraints. Write a UML expression to verify if the time elapsed between a “Question” is being asked online and a “Reply” is being sent back is less than 30 seconds.

- 3.4) Coregions exception on UML extend our abilities to deal with concurrency in sequence diagram. Can you elaborate on that and give an UML example that illustrate difference between “Coregion” and no Coregion.

- 3.5) Is the following figure is allowed in execution graph? If not what it suppose to be; that is, what is the correct form of this figure?



3.6) What is Latency in distributed systems?

3.7) When to use open or closed QNM?

3.8) What is load drivers-Software? When it is recommended to be used?
Hint: It is related to planning performance measurements.

3.9) Principles for Performance-Oriented Design includes 9 principles. One of them is Locality Principle. List without explanation three types of Locality that can be considered during measurement of software performance.

3.10) How Batching Performance Pattern affect the performance of a software system?
Hint: describe the problem and the solution provided by batching pattern.

Question 4: Miscellaneous questions
(20 Points. Partial marks possible)

(5 Points)

- 4.1) Consider an online store that sell fresh meet and fresh produce. Once an order is made, a dispatch wrapping packages is sent. The following algorithm describes the dispatch wrapping packages steps. You need to draw the extended Sequence Diagram. Meat need special packaging different than that of the produce.

```
procedure dispatch
  foreach (order.lineitem)
    if (product.type == meat)
      meat.package
    else
      produce.package
    endif
  endfor
```

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(12 points)

- 4.2) Consider an open QNM with system arrival rate $\lambda = 3$ jobs per second. Following is a list of hardware needed to implement software system which include number of visits for each resource along with mean service time.

Number of visits, V	
CPU	4
Disk	2

Mean service time, S	
CPU	0.01
Disk	0.03

Fill in the table below with the values of metrics for the CPU and Disk; that is, 5 metrics (1-5) are needed for CPU and Disk and 2 metrics are needed for the whole system (metrics 6 and 7).

metrics	CPU	Disk
1. X , throughput		
2. S , mean service time		
3. U , utilization		
4. RT , residence time		
5. N , queue length		
6. Total jobs in system =		
7. System response time =		

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(3 Points)

4.3) Consider a database with an index of 3 levels and an estimated 67% hit rate. Find the estimated number of physical I/Os.

Memory Aid Sheet

Performance metrics based on Execution Profile Charts

T= Total Period, C= Completed jobs, B=Busy Time,

Utilization $U = B/T$

Throughput $X = C/T$

Mean service time $S = B/C$

Number of Jobs $W = \Sigma (\text{\# of jobs})$

Residence time $RT = W/C$

Queue length $N = W/T$

Performance metrics based on Arrival Rate and Average Service Time

Arrival rate, λ

Mean service time, S

Throughput $X = \lambda$

Utilization $U = XS$

Residence time $RT = S/(1-U)$

Queue length $N = X * RT$

Performance metrics based Queueing Network Model

λ System arrival rate

V_i Number of visits to device

S_i Mean service time at device

System throughput, $X_0 = \lambda$

Throughput of device i, $X_i = X_0 \times V_i$

Utilization of device i, $U_i = X_i \times S_i$

Residence time per visit at device i, $RT_i = S_i/(1-U_i)$

Queue length for device i, $N_i = X_i \times RT_i$

System queue length, $N = \Sigma N_i$

System response time, $RT = N/X_0$

Estimate for Random Access of I/O requirements

$$I_p = (i+1) \times (1-h)$$

- I_p is the number of physical I/Os,
- i is the number of levels in the file index, and
- h is an estimate of the hit ratio on the computer system.

Sequential Access Estimate of a File I/O s

number of physical I/Os is

$$n/b$$

- n is the number of records requested and
- b is the block size or number of records that fit into one physical block of data.

Estimate Input Network messages

First consider the amount of data to be transferred in the request (or input). The number of messages for input is

$$M_i = d / (m - h)$$

- d is the amount of input data,
- m is the message size, and
- h is the message header size.

Estimate Output Network messages

The amount of data transferred for the result set.

The number of output messages is

$$M_o = (r \times s) / (m - h)$$

- r is the number of rows in the result set,
- s is the row size,
- m is the message size, and
- h is the message header size.

The performance gain for the refactored solution

$$T_s = M_s \times O$$

- T_s is the processing time saved
- M_s is the number of messages saved
- O is the overhead per message

The cost of dynamic allocation, C , is

$$C = N \times \sum_{depth} (S_c + S_d)$$

- N is the number of calls
- $depth$ is the number of contained objects
- S_c and S_d are the service time to create and to destroy the object, respectively

End